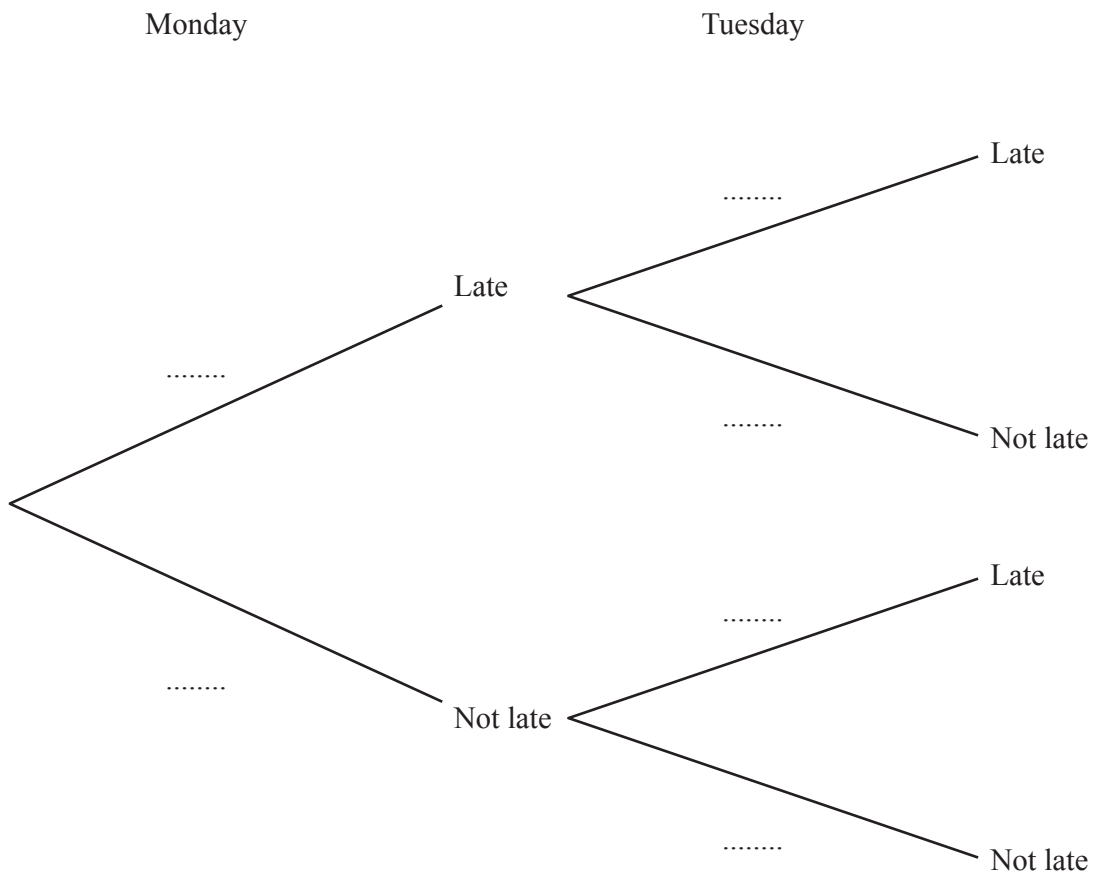
0580_m22_qp_42_Q11

**11** **(a)** The probability that Shalini is late for school on any day is $\frac{1}{6}$.

**(i)** Complete the tree diagram for Monday and Tuesday.




[2]

**(ii)** Calculate the probability that Shalini is late on Monday but is not late on Tuesday.

..... [2]

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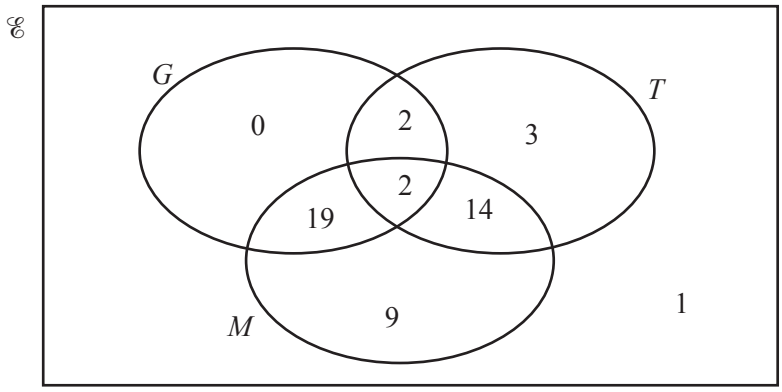
**(b)** The Venn diagram shows the number of students in a group of 50 students who wear glasses ($G$), who wear trainers ($T$) and who have a mobile phone ($M$).




**(i)** Use set notation to describe the region that contains only one student.

..... [1]

**(ii)** Find $n\left(T' \cap \left(G \cup M\right)\right)$.

..... [1]

**(iii)** One student is picked at random from the 50 students.

Find the probability that this student wears trainers but does not wear glasses.

..... [1]

**(iv)** Two students are picked at random from those wearing trainers.

Find the probability that both students have mobile phones.

..... [3]